Lecture 5: Model Limits and Decision Context

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Connection to Lectures 1–4

Lecture 1 defined the forecast target and information available before it. Lecture 2 established chronological evaluation and forecast failure. Lecture 3 examined regularization and refit sensitivity. Lecture 4 fitted nonlinear rules and tested them against simple alternatives. A prediction rule is now available; this lecture asks when it is unsuitable and how to convert its output into a feasible decision with explicit consequences.

The course syllabus places model limits, asymmetric costs and recommendations in Session 5. Prediction intervals and scenarios appear here as decision tools; Lecture 6 develops integrated, rolling evaluation. No separate introductory Python recording is required. The detailed classroom notebook contains complete worked examples 5.1–5.25 with small numbered code steps. Student coding homework is a separate notebook, and every homework task cites its matching public example number.

The main demand file is the same local seed-417 simulation used in Lecture 4. Its later dates have already been inspected; reuse is for instruction, not a new independent test. Other simulations and probability assumptions are labeled explicitly. All figures below are generated by the classroom code, and numerical comparisons are taken from its executed outputs.

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1 Model Boundaries and Operational Safeguards

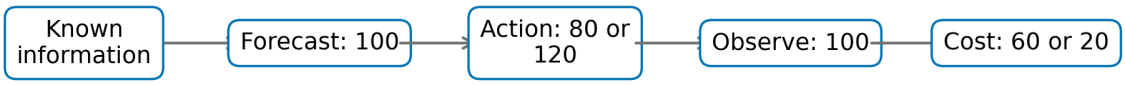
1.1 Information, forecast, action and consequence

Definition: Separate the forecast from the action

A forecast estimates an unknown outcome. An action is a choice made before that outcome is observed. A decision rule maps information and forecasts to an action; a loss records the consequence after the outcome is known. Shortage means demand exceeds stock; excess means stock exceeds demand.

1.1.1 Example 5.1: Separate the forecast from the action

1. Hold the outcome at 100 units and the point forecast at 100.
2. Compare stocking 80 with stocking 120; these are choices, not new forecasts.
3. Charge 3 assumed cost units per shortage unit and 1 per excess unit.
4. The actions incur costs 60 and 20 despite sharing one forecast.



Interpretation and limits. The observed outcome is fixed only to explain how losses are calculated after the event. In a real decision the action is chosen first. A forecast target and an operational action may have the same units, but they are different objects.

Code connection. Lecture code 5.1 in the complete classroom notebook. Separate coding homework HW5.1 uses a related operation with provided starter code and marked blanks.

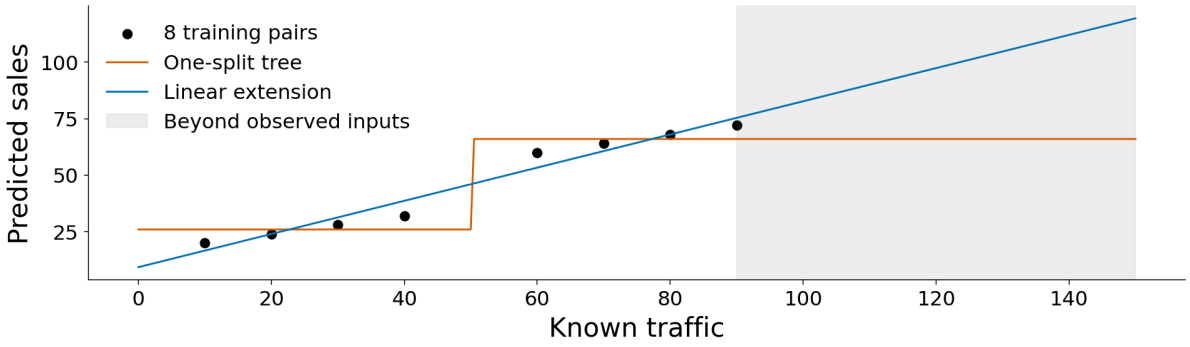
1.2 Extrapolation and the limits of a fitted rule

Definition: A tree does not extend a trend

Extrapolation uses inputs outside the conditions represented in training. A standard mean-valued regression tree returns a stored leaf mean even there; an uncertainty band does not repair a misspecified center.

1.2.1 Example 5.2: A tree does not extend a trend

1. Reuse the eight traffic/sales pairs from lecture code 4.5.
2. Fit the one-split tree and a linear comparison on identical pairs.
3. Extend the query grid beyond the largest training traffic, 90.
4. At traffic 140 the tree still returns 66; the linear extension is an assumption, not verified future truth.



Interpretation and limits. Lecture 4 introduced this structural limit of trees. The linear comparison is not a known correct law outside the observed range either. A wide interval around the wrong center can still miss the future outcome. Support checks and changes in the environment belong beside error scores.

Code connection. Lecture code 5.2 in the complete classroom notebook. Separate coding homework HW5.2 uses a related operation with provided starter code and marked blanks.

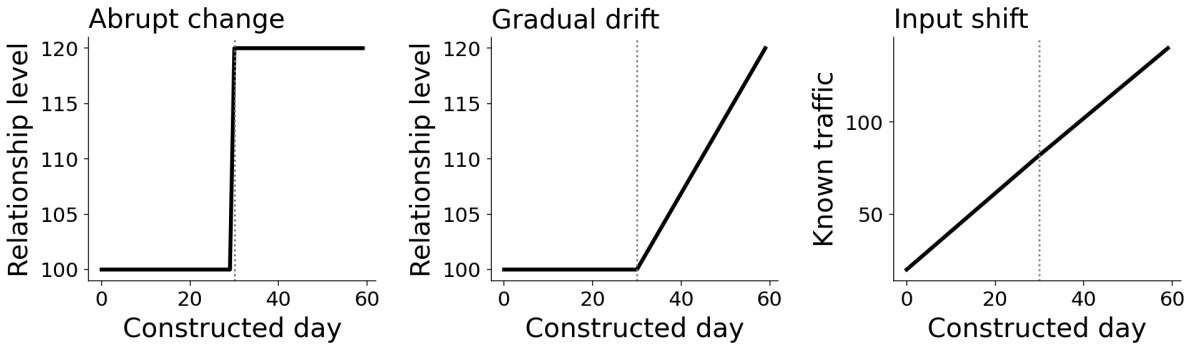
1.3 Different changes in the forecasting environment

Definition: Distinguish three ways the world can change

A regime change alters a relationship abruptly. Parameter drift changes it gradually. An input shift changes the inputs encountered; it need not change the outcome rule itself.

1.3.1 Example 5.3: Distinguish three ways the world can change

1. Use the same horizontal time axis in three separate constructed environments.
2. In the first, the relationship level jumps by 20 starting at day 30.
3. In the second, its level rises gradually starting at day 30.
4. In the third, hold the rule fixed while traffic crosses its earlier maximum.



Interpretation and limits. These panels are three separate constructed environments. A change in demand can reflect changed input values, a changed relationship, or both. Looking at one realized error is not enough to distinguish the explanations. Investigate input definitions and availability before replacing a model.

Code connection. Lecture code 5.3 in the complete classroom notebook. Separate coding homework HW5.3 uses a related operation with provided starter code and marked blanks.

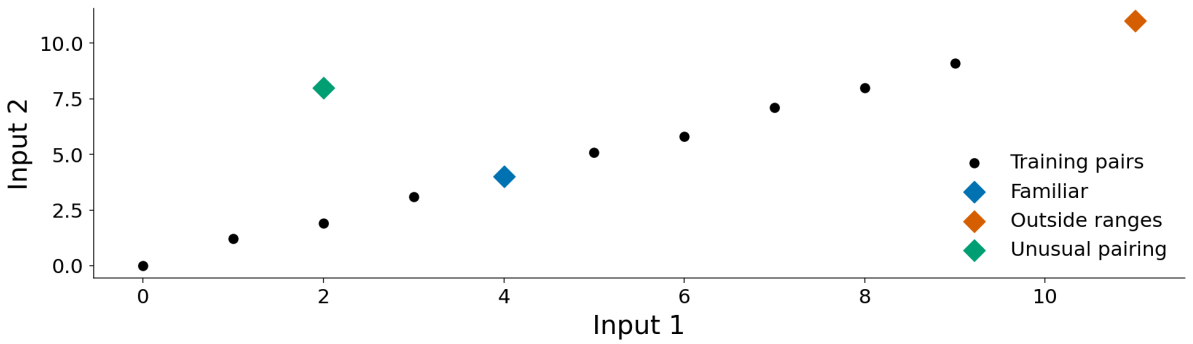
1.4 Observed support and unfamiliar combinations

Definition: Check support without claiming it proves safety

Training support means the input conditions represented by the training observations. A range check can flag obvious departures. Passing each separate range does not guarantee that an input combination was represented.

1.4.1 Example 5.4: Check support without claiming it proves safety

1. Construct paired inputs close to a diagonal band.
2. Compare a familiar pair, an obviously out-of-range pair and an unusual combination inside both marginal ranges.
3. Check coordinate ranges using training observations only.
4. The unusual combination passes both ranges but lies away from the observed pairings.



Interpretation and limits. The coordinate check is a useful warning device, not a sufficient representation of joint support. The point (2,8) combines two individually familiar values in an unfamiliar pairing. This is a concrete reason that passing a simple range check does not establish reliability.

Code connection. Lecture code 5.4 in the complete classroom notebook. Separate coding homework HW5.4 uses a related operation with provided starter code and marked blanks.

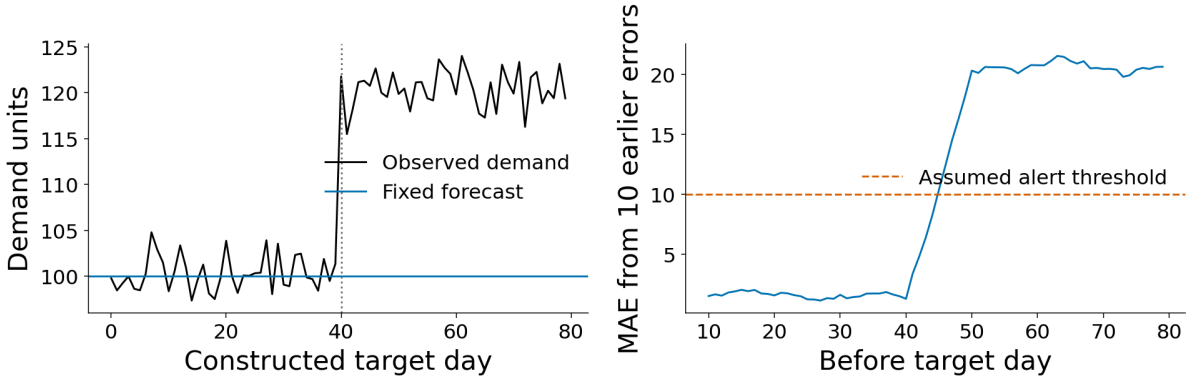
1.5 Evidence arrives after the outcome

Definition: Monitor errors only after outcomes arrive

Operational monitoring checks whether a deployed rule continues to work. A monitoring statistic used before an outcome may include only errors from earlier completed forecasts. An alert is a signal produced when a specified monitoring condition is met.

1.5.1 Example 5.5: Monitor errors only after outcomes arrive

1. Keep the forecast at 100 in a separately labeled simulation.
2. Shift the outcome level starting at day 40.
3. Before each target day, average the absolute errors from the previous 10 completed days.
4. An alert above 10 appears with delay; it does not identify the cause of the change.



Interpretation and limits. The simulated relationship changes at day 40; the first past-error alert occurs at day 45. The statistic deliberately excludes today's error. Alert delay is part of the procedure, not permission to insert today's outcome before forecasting it. The threshold and window were stated in advance for this illustration.

Code connection. Lecture code 5.5 in the complete classroom notebook. Separate coding homework HW5.5 uses a related operation with provided starter code and marked blanks.

1.6 A predefined response to a monitoring signal

Definition: Connect an alert to an explicit fallback rule
A fallback is a previously specified action when the ordinary procedure is unsuitable or its inputs are unavailable. An alert is evidence for review, not proof that an automatic refit solves the problem. Capacity is the largest quantity available under the stated operational plan.

1.6.1 Example 5.6: Connect an alert to an explicit fallback rule

1. Use the alert from example 5.5 rather than looking at the unseen current error.
2. Route an unflagged date to the ordinary decision procedure.
3. Route a flagged date to review, input checks and a predefined capacity plan.

4. Keep the 120-unit emergency capacity assumption separate from the forecast and from estimated model accuracy.



Interpretation and limits. The code saves the ordinary or review procedure and a preset stock plan of 100 or 120 before evaluating the current outcome. Its mean realized cost in this constructed sequence is 7.00 cost units per date. This result describes the specified simulation and rule; it does not establish a generally optimal fallback or prove the change's cause.

Code connection. Lecture code 5.6 in the complete classroom notebook. Separate coding homework HW5.6 uses a related operation with provided starter code and marked blanks.

2 From Error Direction to a Feasible Decision

2.1 Define the sign and units before computing cost

Definition: Keep error direction and business units explicit
With error defined as actual demand minus stocked quantity, a positive error is a shortage and a negative error is excess stock. The sign convention must be fixed before costs are attached.

$$e = y - a$$

Read the expression by meaning.

- y: The realized demand, observed after the decision.
- a: The stock action chosen beforehand.
- e: Their difference; positive means shortage.

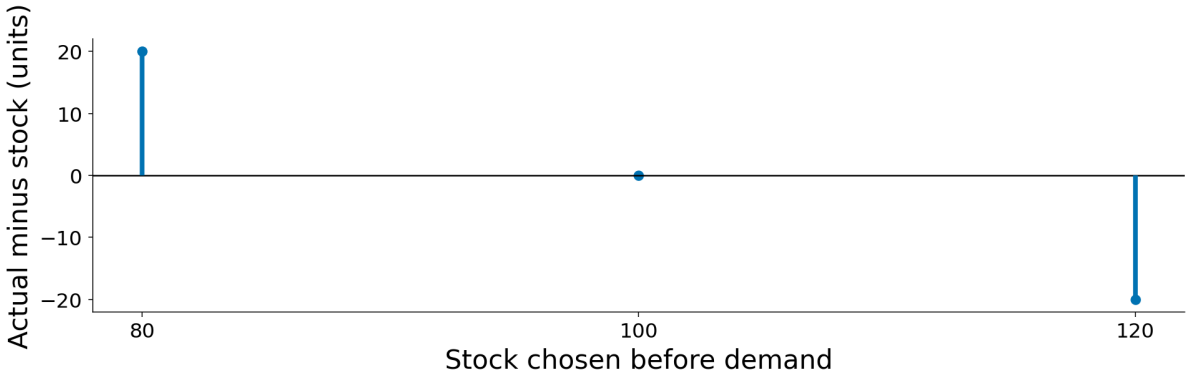
2.1.1 Example 5.7: Keep error direction and business units explicit

1. Use actual demand 100 and candidate stock 80,100,120.

2. Subtract each action from the same actual demand.

3. Separate the positive part of the error from the positive part of its negative.

4. The shortage and excess columns contain nonnegative quantities measured in demand units.



Interpretation and limits. A shortage is the amount by which demand exceeds stock; excess is the amount by which stock exceeds demand. The positive-part operation max(value,0) keeps a quantity when it is positive and otherwise returns zero. Changing the sign convention without changing the coefficients reverses the business meaning.

Code connection. Lecture code 5.7 in the complete classroom notebook. Separate coding homework HW5.7 uses a related operation with provided starter code and marked blanks.

2.2 Charge each type of mistake at its stated rate

Definition: Calculate asymmetric realized loss
Asymmetric loss gives different costs to shortage and excess. A cost coefficient states the cost per unit of a particular mistake; an unexplained number does not create a dollar amount.

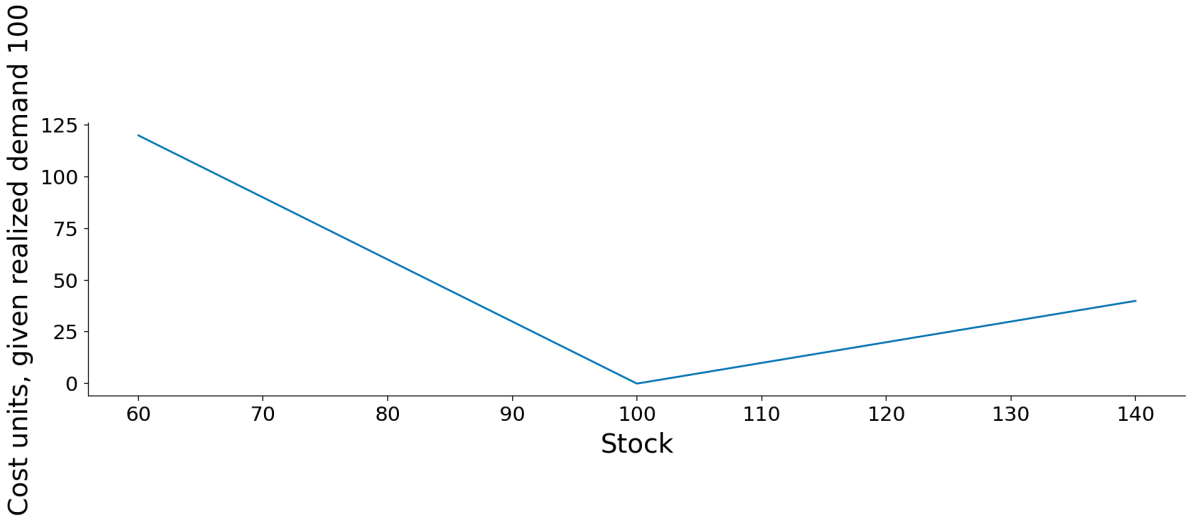
$$L(y, a) = c_u \max(y - a, 0) + c_o \max(a - y, 0)$$

Read the expression by meaning.

- max(y-a,0): The shortage quantity.
- max(a-y,0): The excess quantity.
- c_u and c_o: Assumed cost per unit of shortage and excess.
- L(y,a): The sum of the two cost contributions after demand is observed.

2.2.1 Example 5.8: Calculate asymmetric realized loss

1. Assume a shortage costs 3 cost units per demand unit and excess costs 1.
2. Multiply each nonnegative quantity from 5.7 by its matching coefficient.
3. Add the two cost blocks; only one is positive for a single case.
4. At realized demand 100, stock 80 costs 60 and stock 120 costs 20; this is hindsight loss, not an oracle for choosing tomorrow's action.



Interpretation and limits. The two cost coefficients are assumptions supplied by the business setting. They might represent lost-margin and holding costs if those quantities have been specified, but the classroom numbers are simply cost units per demand unit. A realized loss curve holds the outcome known; it must not be confused with the expected-loss calculation required before the outcome.

Code connection. Lecture code 5.8 in the complete classroom notebook. Separate coding homework HW5.8 uses a related operation with provided starter code and marked blanks.

2.3 Average across unknown possible outcomes

Definition: Choose using expected loss before seeing demand

A probability distribution assigns probability to possible outcomes. Expected loss is a probability-weighted average of the losses an action would produce across those outcomes.

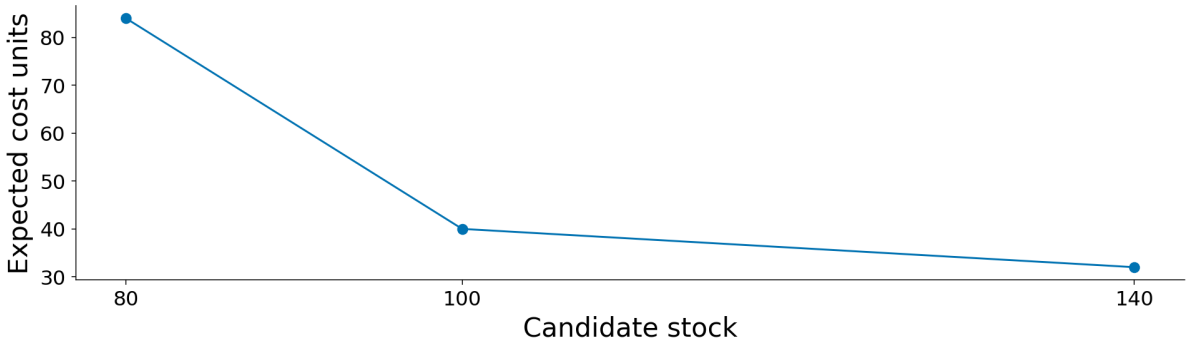
$$\mathbb{E}[L(Y,a)] = \sum_j p_j L(y_j, a)$$

Read the expression by meaning.

- y_j: One possible demand outcome.
- p_j: Its stated probability.
- L(y_j,a): The cost of the same candidate action in that case.
- sum_j: Add the probability-weighted costs over all cases.

2.3.1 Example 5.9: Choose using expected loss before seeing demand

1. Assume demand outcomes 80,100,140 with probabilities .2,.5,.3. These probabilities are stated teaching assumptions.
2. Evaluate each candidate stock against every possible demand.
3. Multiply each loss by its outcome probability and add across outcomes.
4. Among stock 80,100,140, expected costs are 84,40,32; choosing 140 follows the assumed costs and probabilities.



Interpretation and limits. The three probabilities add to one. An action is evaluated against every outcome, not only against the mean. Here the mean demand is 108, but that number alone does not determine the cost-minimizing stock. The stated probability distribution and costs lead to an unconstrained choice of 140 among the displayed candidates.

Code connection. Lecture code 5.9 in the complete classroom notebook. Separate coding homework HW5.9 uses a related operation with provided starter code and marked blanks.

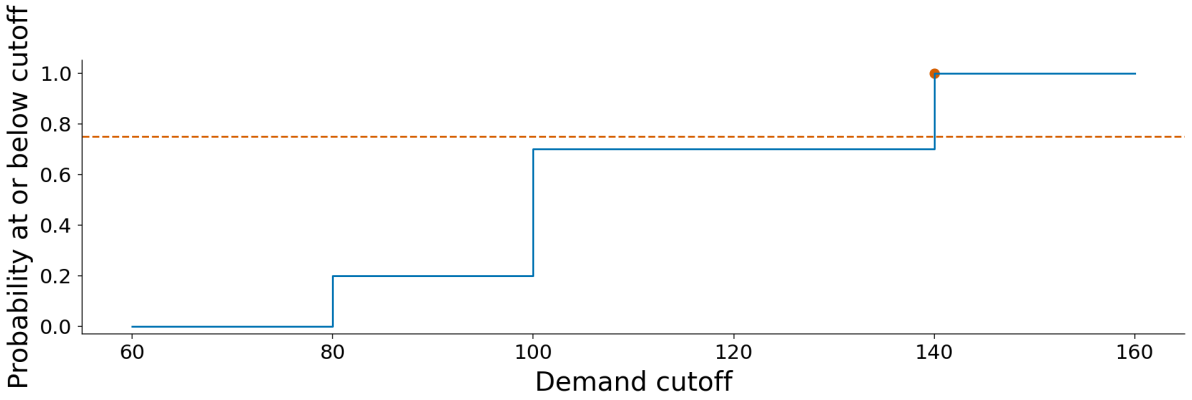
2.4 Cumulative probability and quantiles

Definition: Read a quantile from cumulative probability

A cumulative probability adds the probabilities of outcomes at or below a cutoff. A quantile at level tau is a cutoff reaching that cumulative probability; for this discrete example use the smallest outcome whose cumulative probability is at least tau.

2.4.1 Example 5.10: Read a quantile from cumulative probability

1. Sort the three demand outcomes from low to high.
2. Accumulate their probabilities to .2,.7,1.0.
3. The median is the .5 quantile, equal to 100.
4. The .75 quantile is 140 because the cumulative probability at 100 is only .7.



Interpretation and limits. For the discrete demand distribution, the cumulative mass jumps at each supported value. The smallest-cutoff convention makes the calculation reproducible. An interpolated sample percentile is a different numerical convention; interval examples below state their use of NumPy linear interpolation rather than silently treating the definitions as identical.

Code connection. Lecture code 5.10 in the complete classroom notebook. Separate coding homework HW5.10 uses a related operation with provided starter code and marked blanks.

2.5 Why the cost ratio selects a quantile

Definition: Derive the cost-implied quantile direction

For positive linear shortage and excess costs without additional constraints, a cost-minimizing action is a quantile at level shortage cost divided by the sum of shortage and excess costs. Discrete distributions can have several minimizing actions at a boundary.

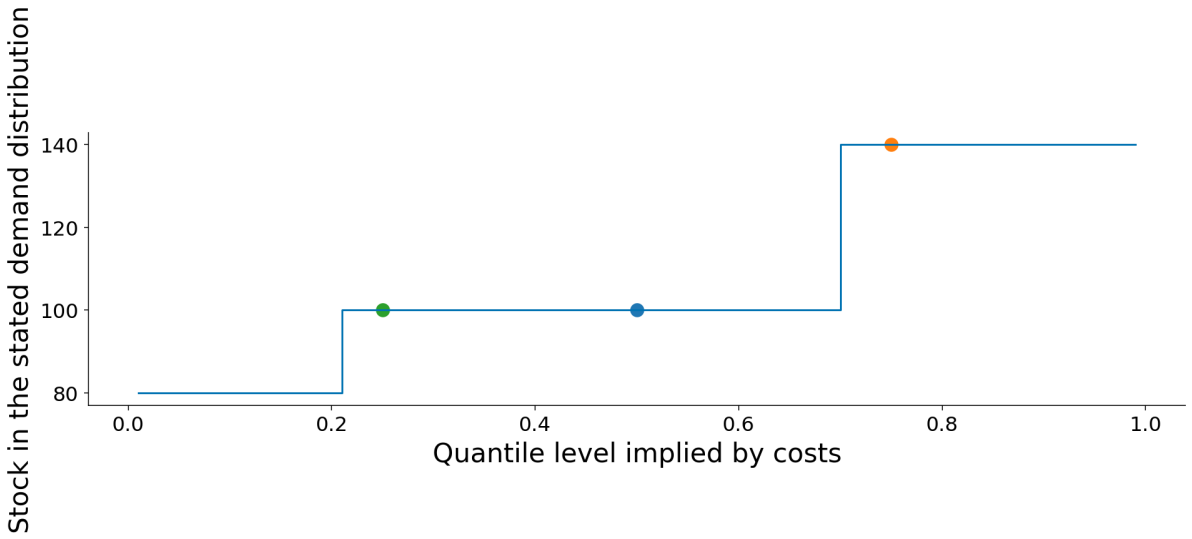
$$\tau^* = \frac{c_u}{c_u + c_o}$$

Read the expression by meaning.

- c_u: Cost per unit of shortage, in the numerator.
- c_u+c_o: Total weight given to the two mistake directions.
- tau*: The quantile level selected by their ratio; it is not the forecast value.

2.5.1 Example 5.11: Derive the cost-implied quantile direction

1. The numerator is the cost of being below demand, not the cost of being above it.
2. Divide shortage cost 3 by total cost weight 4 to obtain .75.
3. Reverse the costs to shortage 1 and excess 3; the quantile level becomes .25.
4. Read the corresponding actions from the same demand distribution: 140 and 100.



Interpretation and limits. Imagine adding one unit of stock. When demand exceeds the current stock, the added unit saves shortage cost. When demand is already below the stock, the unit creates excess cost. For a continuous distribution, probability is spread across

values rather than concentrated at a few isolated outcomes. At a continuous-distribution optimum, expected marginal saving equals expected marginal cost: shortage cost times the upper-tail probability equals excess cost times the lower-tail probability. Rearranging gives the displayed quantile level. In a discrete distribution, the cumulative probability can jump across the target; the smallest crossing is a minimizer, with possible ties at a boundary. Costly overprediction therefore calls for a lower quantile, while costly shortage calls for a higher one.

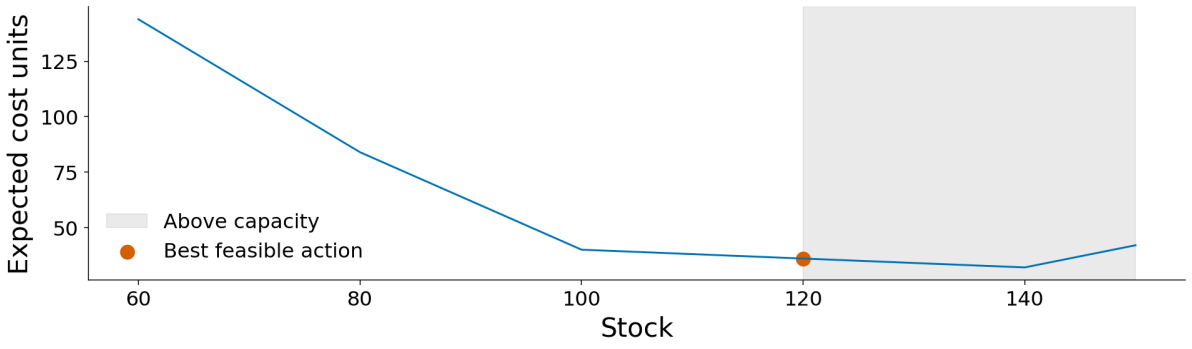
Code connection. Lecture code 5.11 in the complete classroom notebook. Separate coding homework HW5.11 uses a related operation with provided starter code and marked blanks.

2.6 Restrict the action to what is possible

Definition: Impose a feasible action set
A constraint restricts the actions available. The best feasible action minimizes the same expected loss over the allowed set; an unconstrained optimum is not automatically implementable.

2.6.1 Example 5.12: Impose a feasible action set

1. Retain the .75 quantile rule and the three-outcome distribution.
2. Impose a stated capacity of 120 units.
3. Compute expected loss for every integer stock from 60 to 120.
4. The best feasible stock is 120 with expected cost 36; the unconstrained 140-unit action is infeasible.



Interpretation and limits. For this convex, piecewise linear expected loss and a simple upper capacity, clipping the unconstrained quantile at the capacity gives a minimizing feasible action. More complicated action constraints, fixed setup costs, stockout substitution or nonlinear costs require a different decision calculation. Do not attach an arbitrary action percentage to an interval without such a model.

Code connection. Lecture code 5.12 in the complete classroom notebook. Separate coding homework HW5.12 uses a related operation with provided starter code and marked blanks.

3 Accuracy and Uncertainty as Decision Inputs

3.1 A ranking depends on the scoring question

Definition: Separate prediction accuracy from decision loss

Forecast accuracy measures closeness to outcomes under a specified error score. Business loss evaluates the consequence of the action. A lower symmetric error score need not imply lower asymmetric cost.

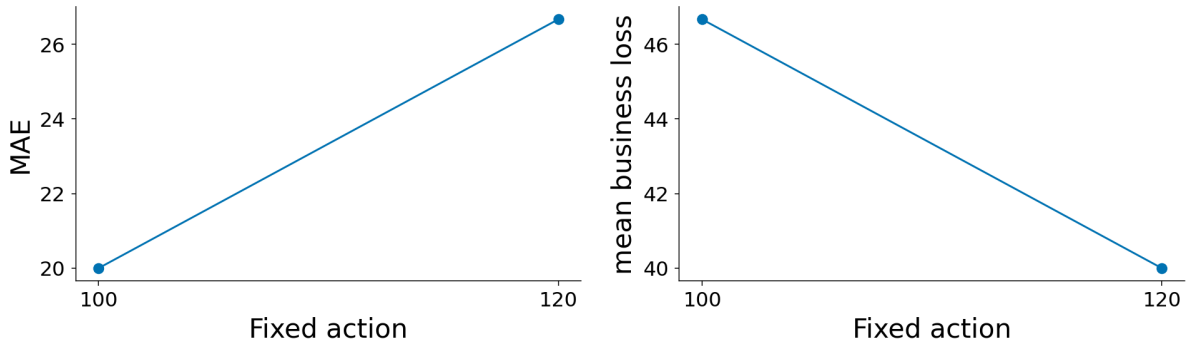
3.1.1 Example 5.13: Separate prediction accuracy from decision loss

1. For this constructed comparison only, let each rule stock its own constant point forecast, 100 or 120, on the same outcomes 80,100,140.

2. Compare constant actions 100 and 120.

3. MAE prefers 100: 20 versus about 26.67.

4. With shortage cost 3 and excess cost 1, average realized cost prefers 120: about 40 versus 46.67. These three equally weighted realized cases are separate from the weighted distribution in 5.9.



Interpretation and limits. The equally weighted three realized cases in this comparison are not the .2/.5/.3 probability model from Example 5.9. The two examples answer different questions and label their weights. Holding cases fixed while changing the loss function demonstrates why model precision and operational usefulness need separate evaluation.

Code connection. Lecture code 5.13 in the complete classroom notebook. Separate coding homework HW5.13 uses a related operation with provided starter code and marked blanks.

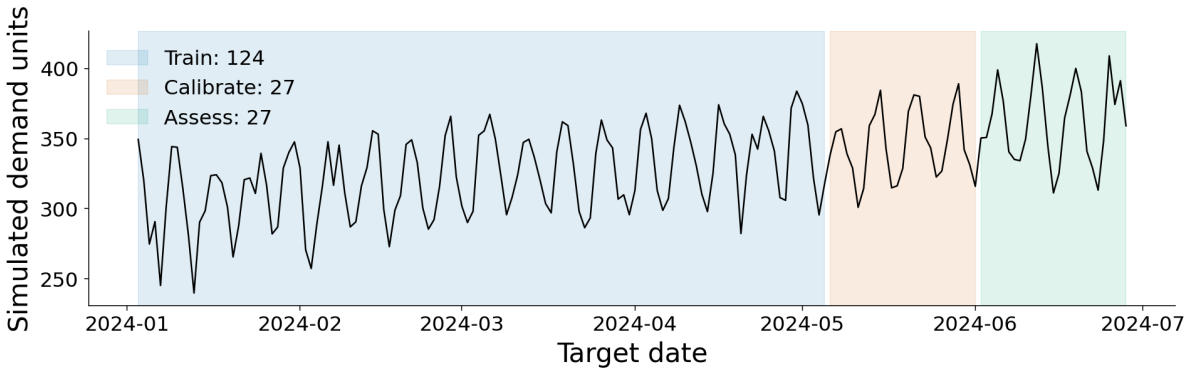
3.2 Train, calibrate and assess the same procedure

Definition: Reuse the classroom data with an honest chronology

A calibration block contains outcomes from earlier forecasts made without those outcomes. It can estimate an error distribution for a fixed prediction procedure. Reusing a previously inspected teaching dataset does not create a new unseen test.

3.2.1 Example 5.14: Reuse the classroom data with an honest chronology

1. Load the same seed-417 classroom demand file used in Lecture 4.
2. Create two known lag inputs; 178 rows remain after the first two dates.
3. Use 124 training rows, 27 calibration rows and 27 later assessment rows.
4. Fit the linear baseline on training only. Keep it fixed during calibration and assessment; unlike Lecture 4's final refit, do not change the model after calibrating its errors.



Interpretation and limits. Lecture 4 found linear, ridge and the selected tree very close on its small validation block; the forest was not a measured champion. This lecture uses the linear baseline as a transparent fixed procedure. Its calibration RMSE is 20.6174, matching Lecture 4 on the same rows. Lecture 4 later refitted settled models using training plus validation; that refitted model is not reused with errors from the earlier fit. The reused assessment data are pedagogical evidence rather than a new prospective test.

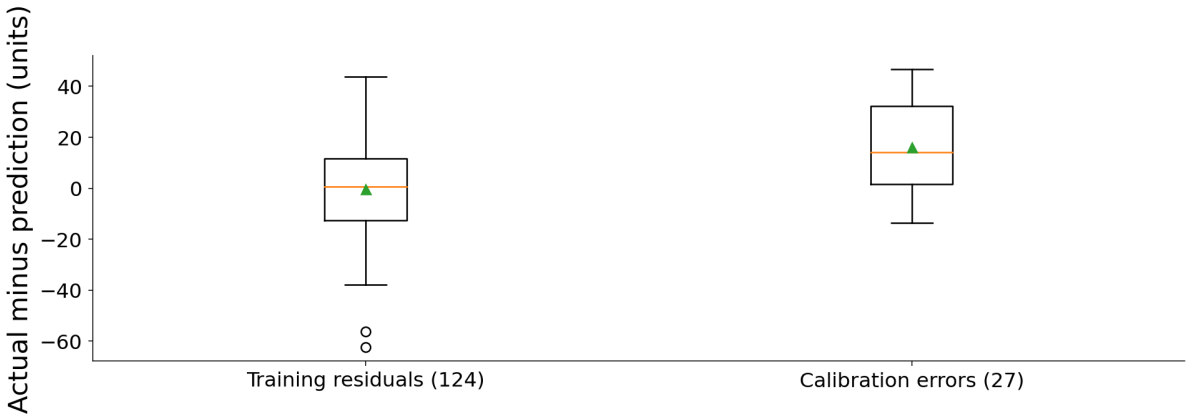
Code connection. Lecture code 5.14 in the complete classroom notebook. Separate coding homework HW5.14 uses a related operation with provided starter code and marked blanks.

3.3 Errors have different information roles

Definition: Distinguish training residuals from later forecast errors
A training residual uses an outcome that helped fit the rule. A later forecast error compares an outcome with a prediction made before seeing that outcome. Their spreads can differ; neither spread is guaranteed to describe a changed future.

3.3.1 Example 5.15: Distinguish training residuals from later forecast errors

1. Fit the specified 100-tree forest on the same 124 training rows.
2. Compute its training residuals and its errors on the 27 calibration dates.
3. Compare the mean absolute error and the 90th percentile of absolute error.
4. Read the measured values; do not assume the ratio or claim a universal size of optimism.



Interpretation and limits. In this run the forest training MAE is 14.65 and calibration MAE is 18.31 demand units. Their 90th percentiles of absolute error are 30.87 and 38.57. This measured difference motivates using later errors, but calibration errors still need to resemble the future task. Estimation error, changes in relationships and unexplained noise can all contribute to forecast error.

Code connection. Lecture code 5.15 in the complete classroom notebook. Separate coding homework HW5.15 uses a related operation with provided starter code and marked blanks.

3.4 Estimate a range without claiming a guarantee

Definition: Construct an empirical prediction interval

A prediction interval gives lower and upper outcome bounds from a stated forecasting procedure. An empirical error quantile is a cutoff estimated from recorded errors. A constant pair of error offsets creates constant-width bands, not state-dependent widening. Nominal coverage is the intended inclusion rate of the procedure; its measured inclusion rate must be checked on outcomes.

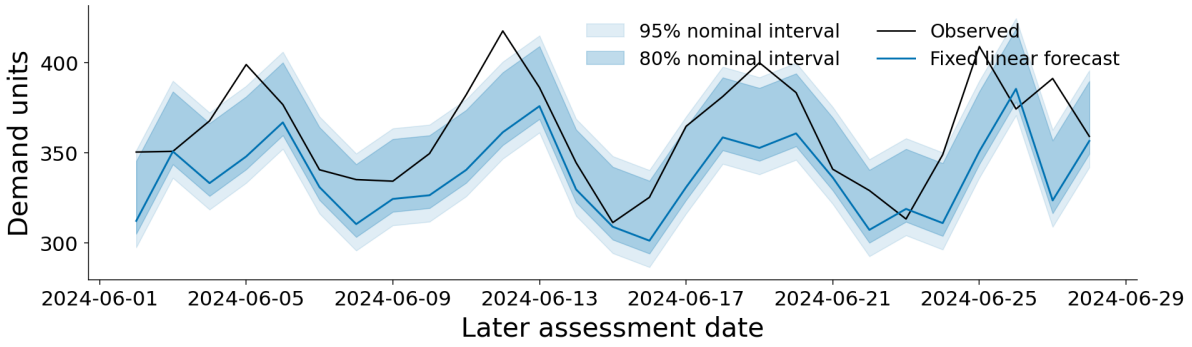
$$[\ell_t, u_t] = [\hat{y}_{t|t-1} + \hat{q}_{\alpha/2},\ \hat{y}_{t|t-1} + \hat{q}_{1-\alpha/2}]$$

Read the expression by meaning.

- target t and origin t-1: The target date and the prior date when its forecast was saved.
- y-hat: The saved point forecast from the unchanged fitted model, made at the prior origin.
- q-hat: A calibration-error cutoff at the stated tail probability.
- alpha: Nominal fraction outside the central interval.
- lower and upper: Add the lower and upper error offsets separately to the same center.

3.4.1 Example 5.16: Construct an empirical prediction interval

1. Use the fixed linear model's 27 calibration errors.
2. For an 80% central interval take the .10 and .90 empirical error quantiles; use NumPy linear interpolation consistently.
3. Add each offset to each later point forecast.
4. For a 95% interval use .025 and .975. These small-sample historical offsets have no automatic time-series coverage guarantee.



Interpretation and limits. The 80% error offsets are -7.18 and 33.21 units. Adding them to every point forecast produces width 40.39 on every date. The unequal offsets retain the measured error asymmetry. These bounds concern future demand outcomes, rather than the fitted average alone. The central interval can fail under distribution change, weak tail information or a changed model. No conditional or finite-sample time-series guarantee is claimed.

Code connection. Lecture code 5.16 in the complete classroom notebook. Separate coding homework HW5.16 uses a related operation with provided starter code and marked blanks.

3.5 Check realized inclusion and useful width

Definition: Measure coverage and width together

Coverage is the fraction of assessed outcomes inside the stated intervals. Width is upper minus lower bound. A target such as 80% is nominal; it is not the observed coverage. Narrowness is useful only with an adequate, honestly evaluated coverage rate.

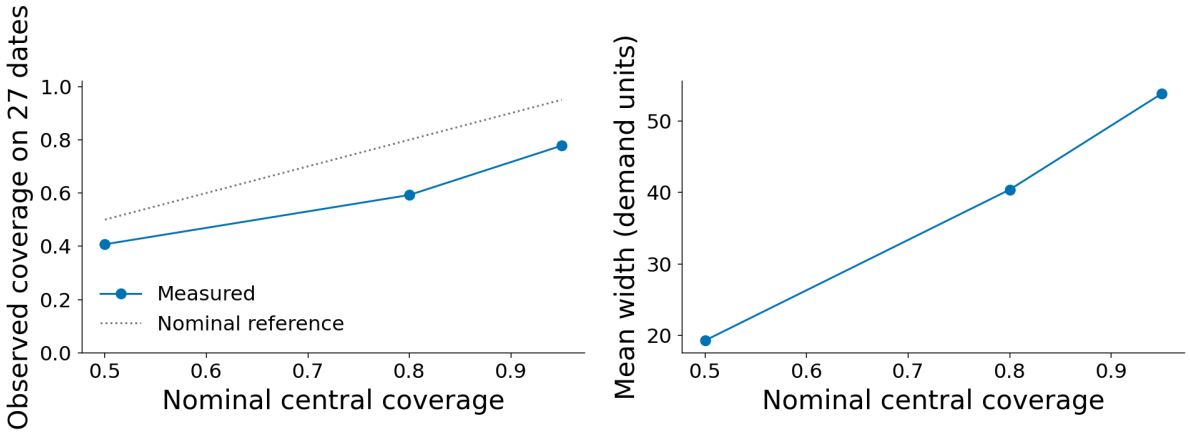
$$\widehat{\text{coverage}} = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}\{\ell_i \le y_i \le u_i\}$$

Read the expression by meaning.

- the indicator: One if a realized outcome is between its saved endpoints, zero otherwise.
- the sum: Number of assessed outcomes inside their intervals.
- n: Number of assessed dates, here 27.
- the ratio: The observed inclusion fraction, not the nominal label.

3.5.1 Example 5.17: Measure coverage and width together

1. Keep the model and calibration errors fixed.
2. Form 50%,80%,95% central intervals using their corresponding quantile offsets.
3. Count inside outcomes on the same 27 later dates and divide by 27.
4. Report width in demand units beside measured coverage. Do not tune these procedures on this already inspected assessment block.



Interpretation and limits. On the 27 reused assessment dates, the nominal 50%, 80% and 95% intervals attain coverage 40.74%, 59.26% and 77.78%. Their widths are 19.31, 40.39 and 53.82 units. Higher nominal coverage enlarges the band, but does not make realized coverage equal the label. Twenty-seven cases are also too few to estimate rare-tail performance precisely. Keep both the observed fractions and their limited sample size visible.

Code connection. Lecture code 5.17 in the complete classroom notebook. Separate coding homework HW5.17 uses a related operation with provided starter code and marked blanks.

3.6 Build state dependence instead of describing it into existence

Definition: Make state-dependent width an explicit method
A state is a condition observed at the forecast origin. State-dependent uncertainty allows forecast-error spread to depend on that condition. To vary width, estimate or model different error distributions; a constant-width band does not do this automatically. Pooled offsets use errors from both states together. Standard deviation measures spread in the same units as the error.

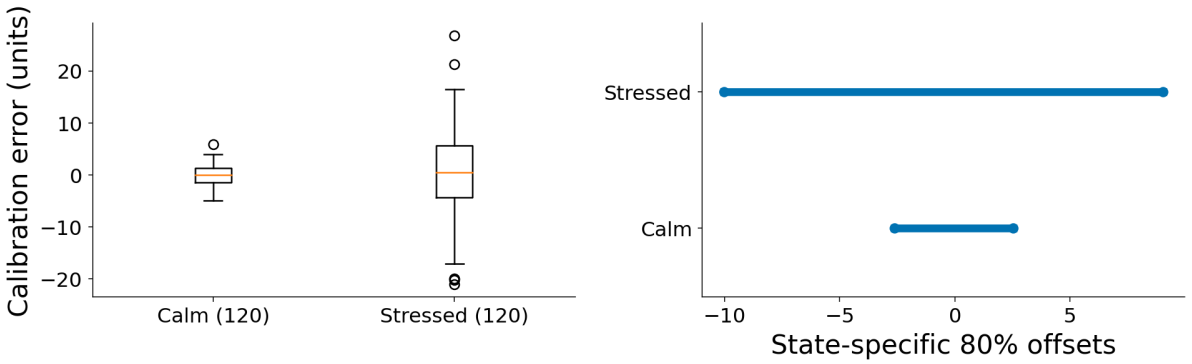
3.6.1 Example 5.18: Make state-dependent width an explicit method

1. In a separate simulation, label each date calm or stressed before its outcome.

2. Generate zero-centered errors with standard deviations 2 and 8; these are constructed assumptions, not market estimates.

3. Estimate .10/.90 offsets from 120 calibration errors in each state.

4. Apply pooled and state-specific offsets to 60 fresh simulated assessment errors per state, then compare coverage and width within states.



Interpretation and limits. The pooled 80% interval has width 11.47 for both states. It covers 100% of the 60 calm outcomes and 41.67% of the 60 stressed outcomes. State-specific widths are 5.15 and 19.05; the corresponding coverage rates are 90% and 76.67%. These measurements come from a separate stated simulation with known pre-outcome state labels. They illustrate a method, not a claim about VIX, market returns or the real demand file. State-specific estimation does not force every finite sample to achieve exactly 80%.

Code connection. Lecture code 5.18 in the complete classroom notebook. Separate coding homework HW5.18 uses a related operation with provided starter code and marked blanks.

4 Scenarios, Thresholds and Policy Comparisons

4.1 Conditional paths and probability claims

Definition: Separate scenarios from assigned probabilities

A scenario specifies a possible future input path and asks what a stated rule implies if those inputs occur. It is conditional analysis. A list of low, middle and high paths does not assign probabilities or create a prediction interval.

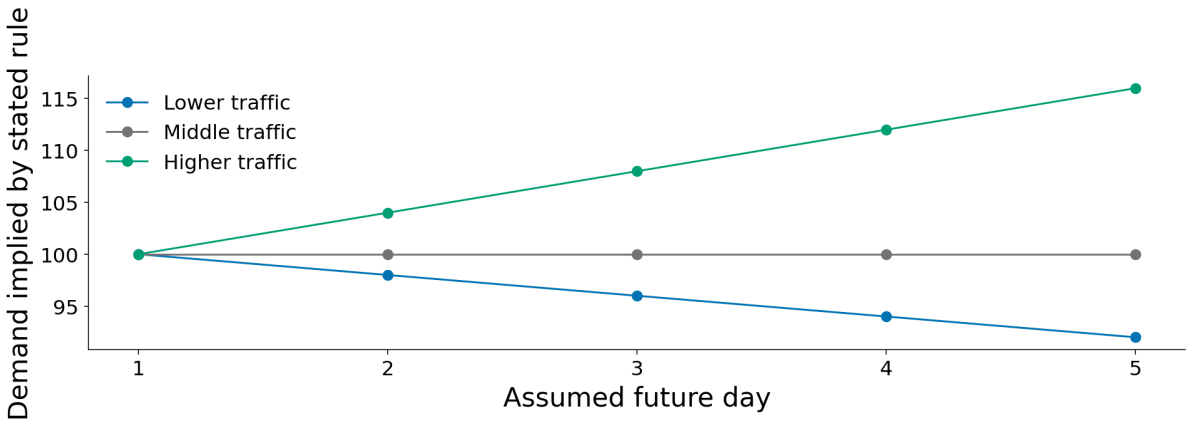
4.1.1 Example 5.19: Separate scenarios from assigned probabilities

1. State a constructed demand rule: baseline 80 plus twice known traffic plus 10 when promotion is active.

2. Specify three future traffic paths for five days; keep promotion off.

3. Pass each path through the same rule.

4. Label the resulting paths as assumed scenarios, not 80% or 95% bounds. Forecasting future inputs is a separate task.



Interpretation and limits. The higher-traffic path reaches demand 116 on day five, the middle path remains at 100, and the lower path reaches 92. Those are outputs of the stated rule and input assumptions. The lines are not assigned probabilities. A scenario envelope can omit plausible outcomes and should not be labeled an 80% prediction interval. Outside a fitted model's support, even carefully written input scenarios can receive unreliable model responses.

Code connection. Lecture code 5.19 in the complete classroom notebook. Separate coding homework HW5.19 uses a related operation with provided starter code and marked blanks.

4.2 Totals require paths, not summed marginal bounds

Definition: Do not add daily interval endpoints into a weekly interval

A joint distribution describes outcomes together, including their dependence. A distribution of weekly total demand requires joint daily paths. Separate daily quantiles alone do not specify that distribution. Dependence means that knowing one outcome can change the probability distribution of another.

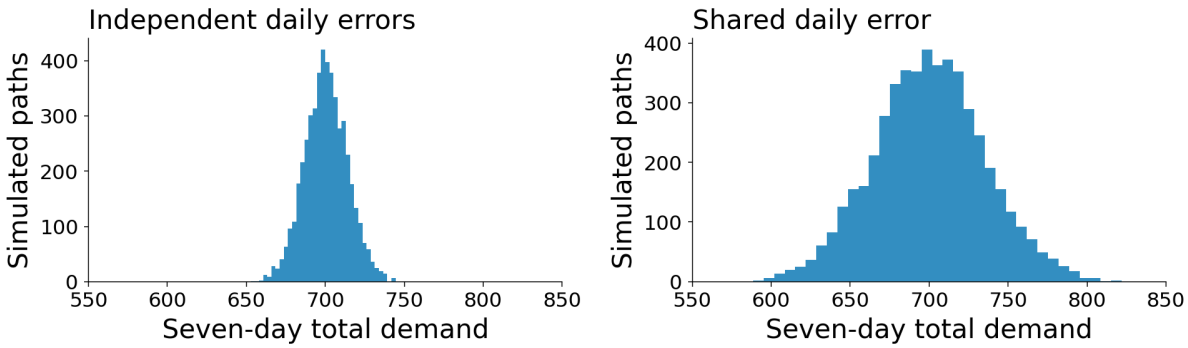
4.2.1 Example 5.20: Do not add daily interval endpoints into a weekly interval

1. Construct 5000 seven-day paths with daily mean demand 100 and error standard deviation 5.

2. In one environment daily errors are independent; in another all seven share the same error.

3. Sum demand within each path and then take the .10/.90 quantiles of totals.

4. The common-error weekly interval is much wider even though every day has the same marginal error distribution.



Interpretation and limits. For 5000 constructed paths per environment, the 80% weekly interval is [683.34, 717.83] under independent errors and [654.14, 745.11] under a shared error. Both have daily error standard deviation 5. The joint dependence changes the total's uncertainty. Adding lower endpoints and adding upper endpoints corresponds to a special dependence assumption, not a universal probability rule. Lecture 6 develops path-based and rolling evaluation further.

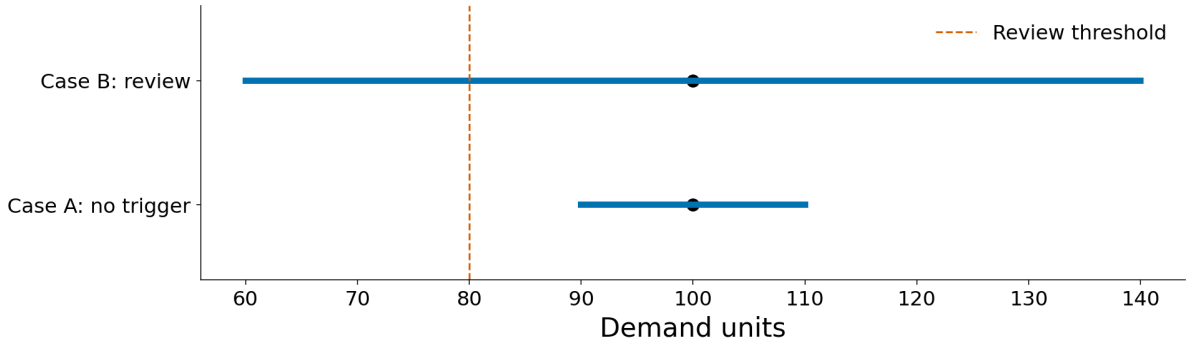
Code connection. Lecture code 5.20 in the complete classroom notebook. Separate coding homework HW5.20 uses a related operation with provided starter code and marked blanks.

4.3 An uncertainty signal and its specified response

Definition: State a threshold rule without inventing an action size
A threshold rule maps a forecast feature to a specified response. A prediction interval can trigger review under such a rule, but it does not by itself determine the economically optimal quantity or the size of a financial position.

4.3.1 Example 5.21: State a threshold rule without inventing an action size

1. Keep the same point forecast, 100, in two constructed cases.
2. Give the cases lower interval bounds 90 and 60.
3. Specify in advance that a lower bound below 80 triggers a capacity-plan review.
4. Only the second case triggers. The rule says review, not an unexplained 20% reduction or a guaranteed loss limit.



Interpretation and limits. The review threshold is 80 demand units. Under the strict below-threshold rule, a lower endpoint exactly at 80 does not trigger. A review plan must specify who checks the input and what feasible response is available. The expected-loss and capacity calculation determines a quantity; an interval trigger supplies a separate reason to review that decision.

Code connection. Lecture code 5.21 in the complete classroom notebook. Separate coding homework HW5.21 uses a related operation with provided starter code and marked blanks.

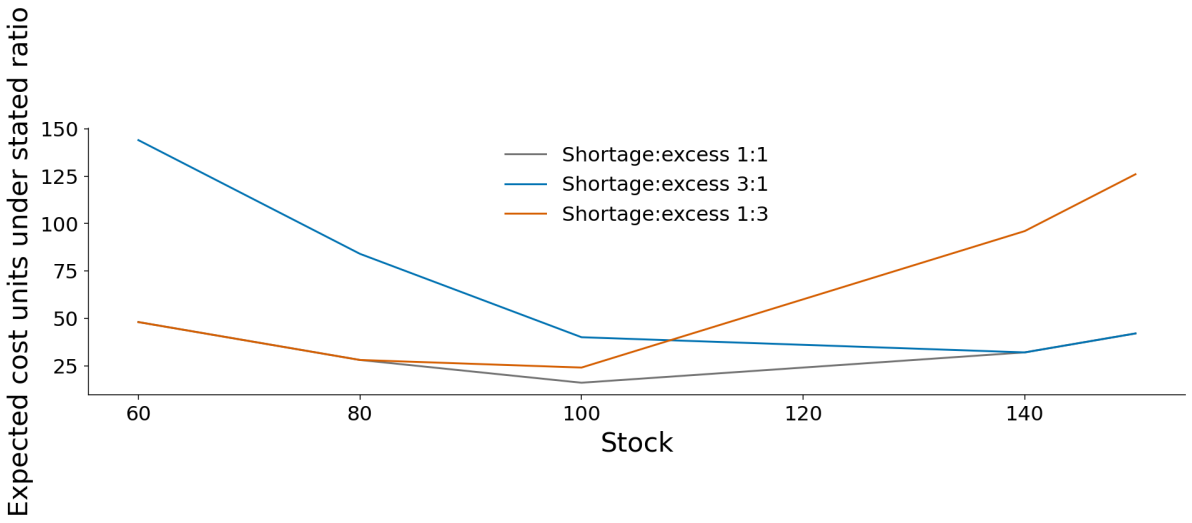
4.4 Expose the assumptions behind the action

Definition: Stress the cost assumptions before recommending

Sensitivity analysis repeats a calculation under explicitly changed assumptions. Robustness means checking how much the conclusion depends on those assumptions; it does not mean a guarantee against every future change.

4.4.1 Example 5.22: Stress the cost assumptions before recommending

1. Keep the same three outcomes and their stated probabilities.
2. Compare shortage/excess cost ratios 1:1,3:1,1:3.
3. Recalculate expected cost over a common action grid for each ratio.
4. Report the chosen actions and what changed. These cost scenarios are assumptions, not facts learned from the demand file.



Interpretation and limits. For the stated three-case distribution, the grid selects stock 100 for costs 1:1, 140 for 3:1, and 100 for 1:3. A lower target quantile need not change the action when the distribution is discrete. Assess which conclusions survive plausible changes in costs, probabilities and capacity. Do not change assumptions solely to make a preferred model appear best.

Code connection. Lecture code 5.22 in the complete classroom notebook. Separate coding homework HW5.22 uses a related operation with provided starter code and marked blanks.

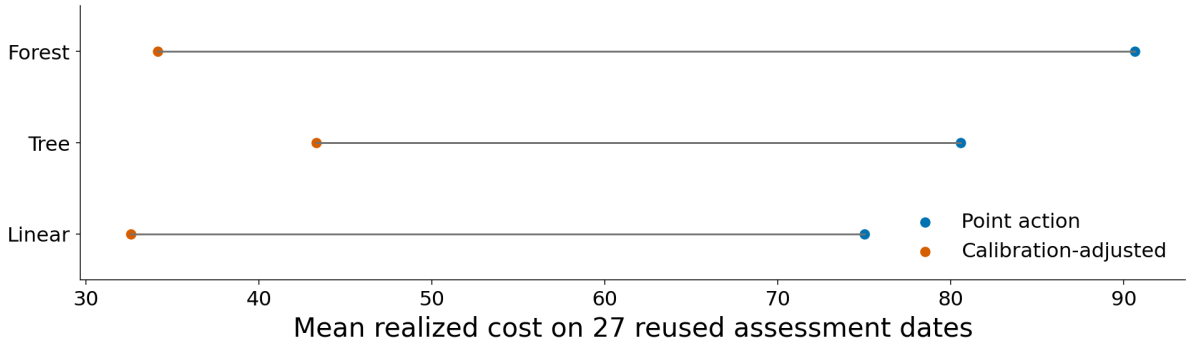
4.5 A calibrated action approximation and its limits

Definition: Evaluate a residual-quantile action adjustment honestly

A conditional quantile is an outcome cutoff given the inputs known at the forecast origin. Adding a calibration-error quantile to a point forecast estimates a cost-sensitive action by assuming that earlier errors remain relevant across future input conditions. It is not automatically the true conditional quantile for every case. The selected tree setting previously used this same block for tuning, so this instructional replay does not claim independent calibration after selection.

4.5.1 Example 5.23: Evaluate a residual-quantile action adjustment honestly

1. Prespecify linear, selected-tree and forest fitting settings from Lecture 4.
2. Fit each model on training only and estimate its .75 error offset from calibration only.
3. Compare unadjusted and adjusted actions on the same 27 later assessment dates.
4. Report measured business loss without choosing a winning policy from this previously inspected block. Do not assume the adjustment must improve realized cost.



Interpretation and limits. The linear, tree and forest point actions have mean realized costs 75.00, 80.58 and 90.66; their calibration-adjusted actions have costs 32.58, 43.32 and 34.14 on this reused block. The .75 error offsets are 25.62, 20.93 and 32.16. These are measured outcomes of three fixed demonstration rules, not a new winner-selection exercise. The tree setting was selected on this same calibration block in Lecture 4, so this replay does not provide independent calibration after tuning. A new study should separate tuning, calibration and later assessment, or explicitly evaluate the reuse. A lower cost-implied quantile means lower than a higher quantile of the same calibrated distribution; it need not lie below an uncalibrated point forecast when calibration errors have positive bias. Varying spread or changed dependence can invalidate the common-offset approximation.

Code connection. Lecture code 5.23 in the complete classroom notebook. Separate coding homework HW5.23 uses a related operation with provided starter code and marked blanks.

5 An Auditable Recommendation and the Next Evaluation Cycle

5.1 A recommendation with traceable arithmetic

Definition: Write a recommendation that can be audited

An auditable recommendation records the information set, forecasting assumptions, available actions, cost rule, uncertainty, chosen action and the condition for review. It separates an assumption from a measured result.

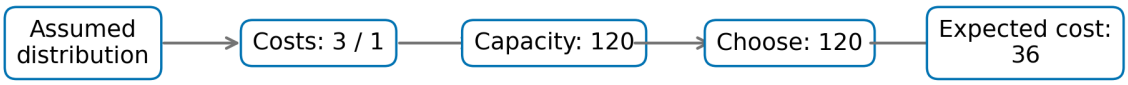
5.1.1 Example 5.24: Write a recommendation that can be audited

1. Reuse the three-outcome decision example; expected demand is 108 and median demand is 100.

2. State costs 3 and 1 and capacity 120.

3. Recommend feasible stock 120 with expected cost 36 under the stated probabilities.

4. Record the limitations: probabilities are assumed, capacity binds, and the realized outcome can differ from the mean.



Interpretation and limits. A useful recommendation states what to do and why, with numbers that can be reconstructed. Here stock 120 is feasible and has expected cost 36 under the assumed distribution. Stock 100 has expected cost 40 under the same assumptions. The difference of 4 is an expected comparison under the specified model, not a guaranteed realized saving. A changed capacity, probability assessment or cost ratio is a reason to revisit the action.

Code connection. Lecture code 5.24 in the complete classroom notebook. Separate coding homework HW5.24 uses a related operation with provided starter code and marked blanks.

5.2 Record the action before attaching the outcome

Definition: Replay decisions before attaching later outcomes
A decision record stores what was known and chosen at the forecast origin. A later evaluation attaches the outcome only after the decision has been saved. This ordering prevents hindsight from silently changing the earlier recommendation.

5.2.1 Example 5.25: Replay decisions before attaching later outcomes

1. Use the fixed linear model and its calibration offset from 5.23.

2. Save target date, prior-day forecast origin, point forecast, interval and planned stock before joining outcomes.

3. Apply a stated capacity of 400 units and nonnegative stock; these are illustrative operational limits.

4. Then attach realized demand and compute cost. Export the record for the integrated rolling evaluation in Lecture 6; do not refit using this assessment outcome.



Interpretation and limits. The exported audit contains 27 decisions. It saves origin and target dates, point forecasts, bounds and capacity-limited stock before attaching observed demand. The mean realized cost is 32.11 under the stated 3:1 costs and 400-unit capacity. The CSV containing decisions before outcomes excludes the outcome and realized-loss columns. The later audit file attaches them for evaluation. Lecture 6 extends this information order into repeated forecast origins, comparisons and communication of uncertainty.

Code connection. Lecture code 5.25 in the complete classroom notebook. Separate coding homework HW5.25 uses a related operation with provided starter code and marked blanks.

6 What a defensible recommendation contains

State the target and forecast origin; identify inputs that were available then; describe the rule and the limits of its observed support. Specify the action set and the cost of each error direction. Distinguish a central forecast, an outcome interval and an assumed scenario. Give an implementable action, its conditional expected consequence and the condition that would trigger review. Preserve the original decision before attaching the realized outcome.

A model can fit complicated patterns and still be unsuitable after a relationship changes. A narrow interval can fail to cover the target. A low RMSE can coexist with a costly action. These are different problems, and the code examples keep their diagnoses and measurements separate.